

 D Y PATIL DEEMED TO BE UNIVERSITY — RAMRAO ADIK — INSTITUTE OF TECHNOLOGY NAVI MUMBAI		B.TECH IN COMPUTER SCIENCE & ENGINEERING (ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)					THIRD YEAR SEM- VI			
Course Code	Course Name	Teaching Scheme (Contact Hours 30)			Credits Assigned					
		Lecture	Practical	Tutorial	Lecture	Tutorial	Prac/Oral	Total		
231CAUCL61	Advanced Machine Learning Lab	--	02	--	--	--	01	01		
		Evaluation Scheme								
		Theory					Term work	Prac./ Oral	Total	
		Internal Assessment			Mid Sem. Exam	End Sem. Exam				Exam duration (Hrs)
		Test 1	Test 2	Avg.						
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• **Course Objectives:**

- To conceptualize the Linear model.
- To understand graphical directed and undirected models.
- To understand Mixture models and EM algorithm.
- To become familiar with Markov Chain and HMM with inference model.

• **Course Outcomes:** After completion of this course, learners will be able to:

- CO1: Identify the generalized linear model and exponential family.
- CO2: Choose an appropriate graphical model for a problem.
- CO3: Identify Mixture model and understand EM algorithm.
- CO4: Apply Markov and Hidden Markov Model in real world problem solution.
- CO5: Analyze and apply the undirected graphical model.
- CO6: Understand and apply Monte Carlo Inference system.

List of Experiments:

Sr. No.	Title of the experiment	CO
1	Build generalized Linear Model.	CO1
2	Develop a regression model for count data using Poisson Regression, a member of the exponential family.	CO1
3	Implement directed Graphical model	CO2
4	Implement Factor Analysis for any real world problem.	CO3
5	Implement EM algorithm for given problem.	CO3
6	Develop a HMM for real world problem (example weather forecasting)	CO4
7	Implement the Viterbi algorithm to find the most likely sequence of hidden states (POS tags) given a sequence of observations (words).	CO4
8	Implement undirected graphical model.	CO5
9	Implement a linear-chain CRF for a sequence labeling task, such as named-entity recognition (NER).	CO5
10	Build Monte Carlo Inference system.	CO6
11	Lab Project	

References:

1. M. P. Deisenroth, A. A. Faisal, C. S. Ong, *Mathematics for Machine Learning*, 1st Edition, Cambridge University Press, MA, USA, 2020.
2. Practical Machine Learning with Tensorflow (nptel.ac.in/courses/106106213).
3. Stanford CS229: Machine Learning Course (www.youtube.com/watch?v=jGwO_UgTS7I).
4. I. Goodfellow, Y. Bengio, and A. Courville, *Deep learning*, 1st Edition, MIT Press, Cambridge, MA, USA, 2016.